Arielle Sedoine Mogoung Notouom

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Professional Summary

Software Engineer with experience designing and deploying cloud-native backend systems in Linux environments. Skilled in Python, distributed systems, REST APIs, WebSockets, Docker, Kubernetes, and cloud platforms (GCP/AWS). Experienced in developing scalable microservices, real-time communication systems, and containerized applications. Strong foundation in software engineering principles including debugging, testing, system design, version control, and collaborative development within agile environments. Passionate about building reliable, high-performance software and continuously learning new technologies.

Professional Experience

Cloud Software Engineer — Vosyn Sep 2024 – Present

- Designed and deployed cloud-native distributed systems for real-time audio/video processing using Python, FastAPI, and WebSockets, achieving end-to-end latency below 200ms for interactive communication workloads.
- Architected and deployed microservices on Google Kubernetes Engine (GKE) and Cloud Run with autoscaling capabilities, enabling the platform to support 100 concurrent users while maintaining reliable performance.
- Developed and deployed 5+ containerized microservices using Docker in Linux-based cloud environments, enabling consistent deployments across Google Kubernetes Engine (GKE) and Cloud Run while supporting scalable application execution through automated container orchestration.
- Designed and implemented a Dynamic Ad Insertion (DAI) system supporting HLS and DASH streaming formats, achieving frame-accurate advertisement placement with less than 1 second insertion delay.

Automation Engineering — InterRent REIT Feb 2024 – Mar 2024

- Supported the integration and deployment of industrial automation control systems across 10+ automated components, improving operational consistency through standardized configuration and validation procedures.
- Performed system validation and troubleshooting activities by analyzing 50+ test scenarios, improving reliability and reducing commissioning issues.
- Contributed to electrical control integration and building automation projects, supporting the deployment, testing, and verification of automated control solutions.

Projects

AI Gateway – Real-Time Audio Streaming Platform

- Developed a Python-based AI Gateway using FastAPI and WebSockets to orchestrate real-time audio streaming between clients and AI processing services.
- Designed asynchronous communication workflows and event-driven architectures to support scalable real-time processing.
- Containerized microservices using Docker and deployed cloud-native workloads on GKE and Cloud Run.

Dynamic Ad Insertion (DAI) System

- Designed and implemented a Dynamic Ad Insertion system using HLS and DASH streaming protocols for seamless advertisement integration.
- Developed manifest manipulation services to dynamically update streaming metadata while maintaining adaptive bitrate playback.
- Implemented distributed media processing workflows to improve streaming reliability and content delivery consistency.

Cloud Video Processing Pipeline (GCP)

- Designed an event-driven video processing pipeline using Google Cloud Storage, Cloud Functions, Pub/Sub, Cloud Run, and Transcoder API.
- Automated transcoding workflows generating multi-resolution video outputs, audio extraction, and DASH streaming assets.
- Integrated monitoring and logging capabilities to improve pipeline observability and operational reliability.

Technical Skills

Programming Languages: Python, Bash, SQL, Object-Oriented Programming (OOP), Data Structures and Algorithms

Backend Development: FastAPI, REST APIs, WebSockets, Microservices Architecture, Asynchronous Programming

Cloud Platforms: Google Cloud Platform (GCP), Cloud Run, Google Kubernetes Engine (GKE), Cloud Functions, Cloud Storage, Pub/Sub, Amazon Web Services (AWS)

Containers & DevOps: Docker, Kubernetes, Git, CI/CD Pipelines, Linux (Ubuntu-based environments)

Distributed Systems & Streaming: Distributed Systems, Event-Driven Architectures, Real-Time Streaming Systems, HLS/DASH Streaming, Media Processing Pipelines, Low-Latency Systems

Education

Master's Degree in Industrial Computing and Automation
EST La Salle — 2018 – 2020

Diploma in Electromechanics of Automated Systems
CFP Jonquière — 2022 – 2024

Languages

- French (Native)
- English (Professional Working Proficiency)